

An elementary proof of a recurrence for OEIS A150500

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Abstract

We give a self-contained derivation of the recurrence recorded as a conjecture by R. J. Mathar in OEIS A150500. One of the coordinate constraints is redundant, and the walk count is a binomial transform of squared one-dimensional meander counts. A second-order differential equation for the series of squared central binomial coefficients then yields the stated recurrence by coefficient extraction. The argument also recovers the generating-function and sequence relations already recorded in OEIS.

Let a_n be the number of length- n walks starting at $(0, 0, 0)$, remaining in $\mathbb{Z}_{\geq 0}^3$ at every step, and using the step set

$$S = \{(-1, -1, -1), (-1, -1, 1), (0, 1, 0), (1, 1, -1), (1, 1, 1)\}.$$

These numbers form OEIS A150500 [1]. The recurrence below is listed there as a conjecture attributed to R. J. Mathar, dated October 29, 2025. The generating function for the auxiliary sequence A018224 and the relation to A201805 are already recorded in OEIS [2, 1, 3]. For background on hypergeometric generating functions for quarter-plane walks, see [4]. This note derives the stated recurrence directly from the walk definition, without assuming an auxiliary recurrence.

Theorem 1. *For every integer $n \geq 4$,*

$$\begin{aligned} (n+1)^2 a_n - (9n+4)a_{n-1} + (-22n^2 + 27n + 15)a_{n-2} \\ - 3(n-2)(8n-31)a_{n-3} + 45(n-2)(n-3)a_{n-4} = 0. \end{aligned} \tag{1}$$

The initial values are $a_0 = 1$, $a_1 = 2$, $a_2 = 7$, and $a_3 = 25$.

Proof. We use the convention $\binom{n}{j} = 0$ if $j < 0$ or $j > n$, for $n \geq 0$. All generating functions are formal power series over $\mathbb{Q}$, and derivatives are formal derivatives. Intermediate expressions containing x^{-1} or z^{-1} are interpreted in the corresponding formal Laurent series fields; the expressions used as generating functions have no negative-degree terms.

Counting the walks. At every prefix of a walk, $y - x$ equals the number of occurrences of the step $(0, 1, 0)$. Consequently, $x \geq 0$ implies $y \geq 0$. Call $(0, 1, 0)$ the special step. After deleting all special steps, the remaining steps project onto the four steps $(\pm 1, \pm 1)$ in the (x, z) -plane. Each such step corresponds to exactly one of the four nonspecial steps in S . Thus a projected walk is equivalent to an ordered pair of one-dimensional walks of the same length, each starting at zero, having increments ± 1 , and remaining nonnegative at every prefix.

Let M_m count these one-dimensional walks of length m , with unrestricted endpoint. A walk ending at $m - 2j \geq 0$ has j downward steps, so there are $\binom{m}{j}$ unrestricted walks to that

endpoint. Among these, a walk that reaches -1 is transformed by reversing every increment up to and including its first visit to -1 . This is a bijection onto the unrestricted walks ending at $m - 2j + 2$, which have $j - 1$ downward steps. The inverse reverses the increments up to the first visit to 1 . Hence the number of nonnegative walks ending at $m - 2j$ is $\binom{m}{j} - \binom{m}{j-1}$. Summing gives

$$M_m = \sum_{j=0}^{\lfloor m/2 \rfloor} \left(\binom{m}{j} - \binom{m}{j-1} \right) = \binom{m}{\lfloor m/2 \rfloor}.$$

There are M_m^2 ordered pairs of such coordinate walks. Inserting $n-m$ special steps at any chosen positions preserves the nonnegativity of x and z , and $y = x + (\text{number of special steps so far})$ remains nonnegative as well. Deletion and insertion are inverse constructions, so

$$a_n = \sum_{m=0}^n \binom{n}{m} M_m^2. \quad (2)$$

Generating functions. Put

$$C(z) = \sum_{m \geq 0} M_m^2 z^m, \quad D(z) = \sum_{k \geq 0} \binom{2k}{k}^2 z^{2k}, \quad A(x) = \sum_{n \geq 0} a_n x^n.$$

Since $\binom{2k+2}{k+1} = 2\binom{2k+1}{k}$, the even and odd parts of C give

$$C(z) = D(z) + \frac{D(z) - 1}{4z}. \quad (3)$$

The quotient is a power series because $D(z) - 1$ is divisible by z^2 . Using $\sum_{n \geq m} \binom{n}{m} x^n = x^m / (1-x)^{m+1}$ in (2), and writing $u = x/(1-x)$, we obtain

$$A(x) = \frac{1}{1-x} C(u) = \frac{1+3x}{4x(1-x)} D(u) - \frac{1}{4x}.$$

The substitution is valid formally because u has zero constant term. The apparent negative-degree terms cancel, as is also clear from the first expression for A . Define $B(x) = (1-x)^{-1} D(u)$. Rearranging gives

$$B(x) = \frac{4xA(x) + 1}{1+3x}. \quad (4)$$

A differential equation. Write $D(z) = \sum_{m \geq 0} d_m z^m$, with $d_{-1} = d_{-2} = 0$. For even $m \geq 2$, factorial cancellation gives

$$m^2 d_m = 16(m-1)^2 d_{m-2}.$$

For odd m , both coefficients vanish, and the identity also holds at $m = 0$. Therefore, for $E = z d/dz$,

$$(E^2 - 16z^2(E+1)^2) D(z) = 0. \quad (5)$$

Set $\theta = x d/dx$ and define

$$Q = (1 - 3x - 13x^2 + 15x^3)\theta^2 + (-3x - 26x^2 + 45x^3)\theta + (-x - 13x^2 + 30x^3). \quad (6)$$

The change of variable $z = u = x/(1-x)$ turns E into $T = x(1-x)d/dx$. For $H = (1-x)B = D(u)$, the product rule gives

$$T^2 H - \frac{16x^2}{(1-x)^2} (T^2 H + 2TH + H) = QB. \quad (7)$$

Equations (5) and (7) imply $QB = 0$.

Eliminating the auxiliary series. Define

$$\begin{aligned} L = & (1 - 22x^2 - 24x^3 + 45x^4)\theta^2 \\ & + (2 - 9x - 61x^2 - 3x^3 + 135x^4)\theta \\ & + (1 - 13x - 19x^2 + 21x^3 + 90x^4). \end{aligned} \quad (8)$$

For every $F \in \mathbb{Q}[[x]]$, the product rule yields

$$Q\left(\frac{4xF + 1}{1 + 3x}\right) = \frac{4x}{(1 + 3x)^2}(LF - (1 - 5x)). \quad (9)$$

To verify this identity, put $h = 4x/(1 + 3x)$, $g = 1/(1 + 3x)$, and write $Q = q_2\theta^2 + q_1\theta + q_0$, with coefficients from (6). Then

$$\begin{aligned} Q(hF + g) = & q_2h\theta^2F + (2q_2\theta h + q_1h)\theta F \\ & + (q_2\theta^2h + q_1\theta h + q_0h)F + Qg. \end{aligned}$$

Here $\theta h = 4x/(1 + 3x)^2$ and $\theta^2h = 4x(1 - 3x)/(1 + 3x)^3$; substitution gives (9). Taking $F = A$ and using (4) and $QB = 0$ gives

$$\frac{4x}{(1 + 3x)^2}(LA - (1 - 5x)) = 0.$$

The prefactor is a nonzero element of the integral domain $\mathbb{Q}[[x]]$, so

$$LA = 1 - 5x. \quad (10)$$

For coefficient extraction, rewrite (8) as

$$\begin{aligned} L = & (\theta + 1)^2 + x(-9\theta - 13) + x^2(-22\theta^2 - 61\theta - 19) \\ & + x^3(-24\theta^2 - 3\theta + 21) + 45x^4(\theta + 1)(\theta + 2). \end{aligned}$$

For any polynomial p and $n \geq j$, $[x^n]x^j p(\theta)A = p(n - j)a_{n-j}$. Thus, for $n \geq 4$, the coefficient of x^n in (10) is precisely (1). The initial values follow from (2): $a_0 = 1$, $a_1 = 1 + 1 = 2$, $a_2 = 1 + 2 + 4 = 7$, and $a_3 = 1 + 3 + 12 + 9 = 25$. $\square$

Relation to previously recorded formulas. Writing $B(x) = \sum_{n \geq 0} b_n x^n$ and expanding its definition gives

$$b_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{2k}{k}^2.$$

This is the formula for A201805 recorded by P.-L. Giscard in 2014 [3]. Extracting the coefficient of x^{n+1} from $(1 + 3x)B = 4xA + 1$ gives $4a_n = b_{n+1} + 3b_n$ for $n \geq 0$, the relation recorded by M. van Hoeij in 2024 [1]. Also, $D(z) = {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 16z^2)$, so (3) recovers the generating function for A018224 recorded by M. van Hoeij in 2009 [2]. These formulas are included to identify the connection with existing entries; the proof above derives them from the walk count and does not assume a conjectured auxiliary recurrence.

References

- [1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A150500. Recurrence recorded by R. J. Mathar, October 29, 2025; relation to A201805 recorded by M. van Hoeij, November 29, 2024. <https://oeis.org/A150500>. Accessed October 5, 2026.
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